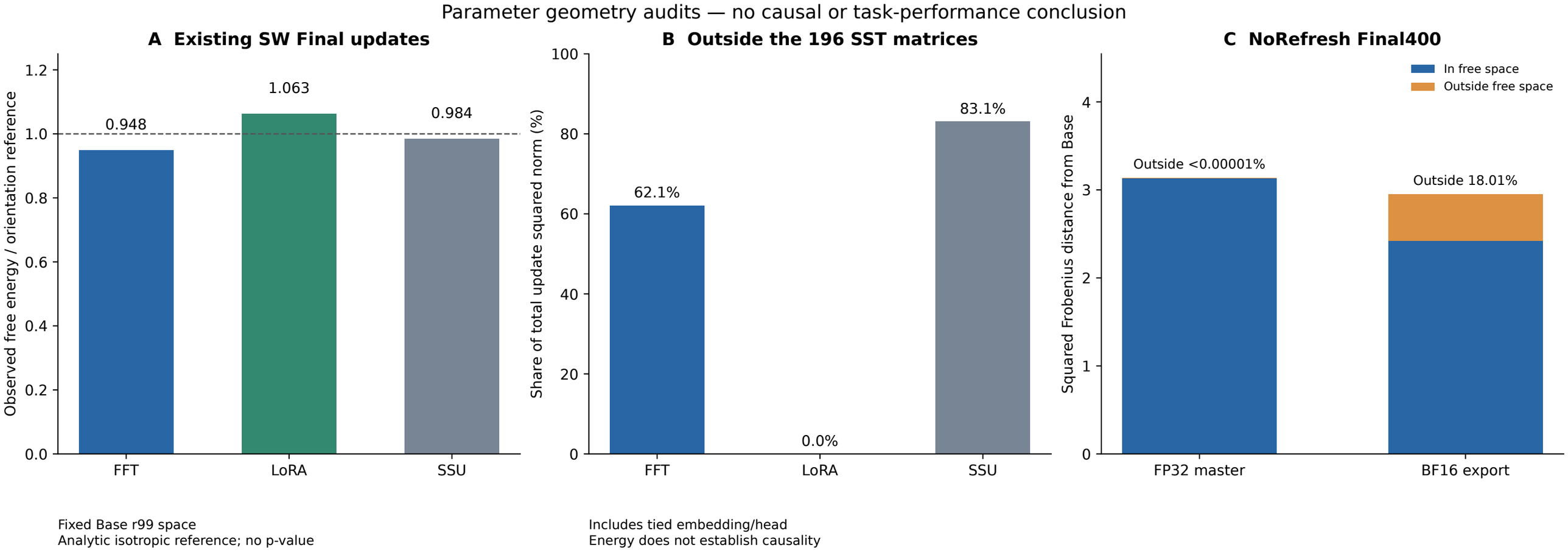
Parameter geometry audits — no causal or task-performance conclusion
A Existing SW Final updates
Observed free energy / orientation reference
0.948
1.063
0.984
FFT
LoRA
SSU
Fixed Base r99 space
Analytic isotropic reference; no p-value
B Outside the 196 SST matrices
Share of total update squared norm (%)
62.1%
0.0%
83.1%
FFT
LoRA
SSU
Includes tied embedding/head
Energy does not establish causality
C NoRefresh Final400
Squared Frobenius distance from Base
In free space
Outside free space
Outside <0.00001%
Outside 18.01%
FP32 master
BF16 export